

ECEN 428/722 - Post Lab Report

Lab Number: 7

Lab Title: Kalman Filter

Section Number: 602

Student's Name: Sai Vikhyat Pattar

Student's UIN: 936001854

Date: 11/6/2025

TA: Cristhian Roman Vicharra

Objective:

The objective of this lab is to design and implement a **Kalman Filter** on FPGA for real-time state estimation. The filter predicts and updates the system's position and velocity in the presence of noise using recursive equations. The design applies fixed-point arithmetic and matrix-based computation in Verilog to demonstrate efficient hardware realization of digital estimation systems.

Methodology:

1. Extracted and opened the **lab7_base.zip** project in Vivado.
2. Added the required Verilog source files — **kalman.v** and **top_kalman.v** — along with the necessary block memory IPs.
3. Implemented the Kalman filter architecture using **16-bit signed fixed-point arithmetic** with 10 fractional bits to balance accuracy and hardware efficiency.
4. Defined the system matrices (F, B, H, Q, and R) and initialized all variables for state estimation.
5. Designed the filter to perform both the **prediction** and **update** steps sequentially within each clock cycle.
6. Configured multiple **block RAMs** to store intermediate results and matrices for efficient data access.
7. Verified functional correctness using **behavioral simulation** in Vivado to ensure proper convergence of estimated states.
8. Generated the **bitstream**, exported the hardware design to **Vitis**, and created the BOOT.bin file.
9. Loaded the bitstream onto the **Zybo board** and executed the **lab7_kalman_test** program to test real-time FPGA operation.
10. Recorded terminal outputs and analyzed performance in terms of timing, accuracy, and hardware resource utilization.

Design:

The implemented design follows the standard Kalman filter structure with:

- **Predict Stage:** Computes predicted state (x_{predict}) and covariance (P_{predict}) based on previous estimates and control input.
- **Update Stage:** Corrects predictions using new measurements to reduce estimation error.
- **Fixed-Point Arithmetic:** All multiplications and additions use 16-bit signed values with 10 fractional bits for efficient FPGA resource utilization.
- **Control Logic:** Ensures proper sequencing between matrix operations with `outen` flag asserted only during valid output cycles.
- **RAM Blocks:** Store and retrieve matrices and intermediate results efficiently using parallel read/write ports.

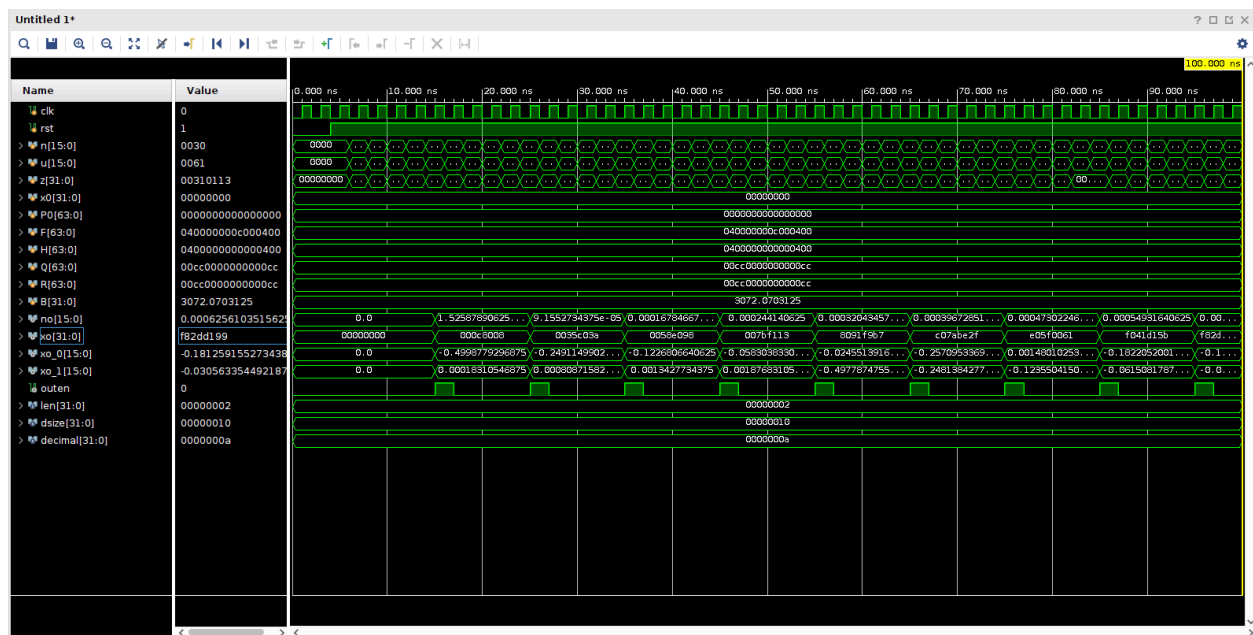
Implementation:

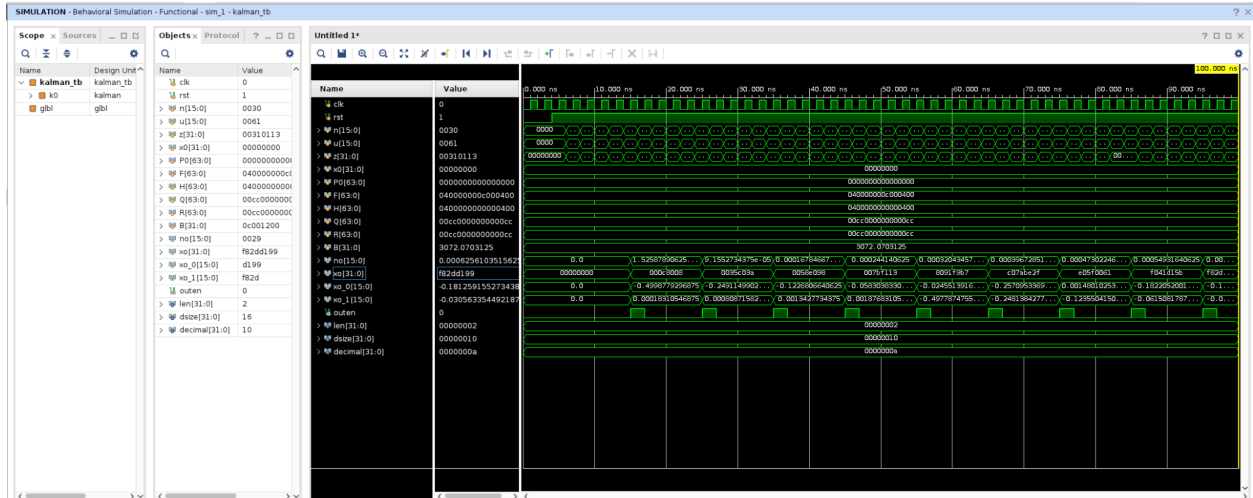
The design was successfully synthesized and implemented in **Vivado**, and the FPGA execution was verified on the Zybo board.

- **Block Memory Configuration:**
 - blk_mem_gen_0 → True Dual-Port RAM (8-bit × 20,000-depth)
 - blk_mem_gen_1 → Single-Port RAM (96-bit × 1,000-depth)
 - blk_mem_gen_2 → Single-Port RAM (32-bit × 1,000-depth)
- **Simulation:** Behavioral simulation confirmed correct recursive estimation behavior and stable convergence.
- **Hardware Verification:** The program ./lab7_kalman_test was executed successfully, showing real-time estimation results on the terminal with absolute error < 1.5, indicating accurate FPGA-based filtering.

Results:

Behavioral simulation:





Terminal output:

```
time: 97.700000, filtered data: 7.081055, difference with ideal output: 0.886719
time: 98.500000, filtered data: 7.144531, difference with ideal output: 0.853516
zynq> ./lab7 kalman_test 5
Your interval between sampling is 5
time: 0.100000, filtered data: 0.061523, difference with ideal output: 0.061523
time: 0.900000, filtered data: 0.121094, difference with ideal output: 0.101562
time: 1.700000, filtered data: 0.124023, difference with ideal output: 0.029297
time: 2.500000, filtered data: 0.176758, difference with ideal output: -0.015625
time: 3.300000, filtered data: 0.286133, difference with ideal output: 0.039062
time: 4.100000, filtered data: 0.341797, difference with ideal output: 0.087891
time: 4.900000, filtered data: 0.375000, difference with ideal output: 0.101562
time: 5.700000, filtered data: 0.431641, difference with ideal output: 0.082031
time: 6.500000, filtered data: 0.500000, difference with ideal output: 0.053711
time: 7.300000, filtered data: 0.613281, difference with ideal output: 0.112305
time: 8.100000, filtered data: 0.677734, difference with ideal output: 0.169922
time: 8.900000, filtered data: 0.697266, difference with ideal output: 0.169922
time: 9.700000, filtered data: 0.728516, difference with ideal output: 0.125000
time: 10.500000, filtered data: 0.817383, difference with ideal output: 0.117188
time: 11.300000, filtered data: 0.899414, difference with ideal output: 0.144531
time: 12.100000, filtered data: 0.929688, difference with ideal output: 0.167969
time: 12.900000, filtered data: 0.997070, difference with ideal output: 0.214844
time: 13.700000, filtered data: 1.022461, difference with ideal output: 0.165939
time: 14.500000, filtered data: 1.102539, difference with ideal output: 0.147461
time: 15.300000, filtered data: 1.208984, difference with ideal output: 0.200195
time: 16.100000, filtered data: 1.246094, difference with ideal output: 0.230469
time: 16.900000, filtered data: 1.315430, difference with ideal output: 0.279297
time: 17.700000, filtered data: 1.352539, difference with ideal output: 0.241211
time: 18.500000, filtered data: 1.424805, difference with ideal output: 0.215820
time: 19.300000, filtered data: 1.496094, difference with ideal output: 0.232422
time: 20.100000, filtered data: 1.505859, difference with ideal output: 0.235352
time: 20.900000, filtered data: 1.534180, difference with ideal output: 0.244141
time: 21.700000, filtered data: 1.576172, difference with ideal output: 0.209961
time: 22.500000, filtered data: 1.652344, difference with ideal output: 0.189453
time: 23.300000, filtered data: 1.716797, difference with ideal output: 0.199219
time: 24.100000, filtered data: 1.762695, difference with ideal output: 0.238281
time: 24.900000, filtered data: 1.811523, difference with ideal output: 0.267578
time: 25.700000, filtered data: 1.847656, difference with ideal output: 0.227539
time: 26.500000, filtered data: 1.919922, difference with ideal output: 0.203125
time: 27.300000, filtered data: 1.999023, difference with ideal output: 0.227539
time: 28.100000, filtered data: 2.026367, difference with ideal output: 0.248047
time: 28.900000, filtered data: 2.041016, difference with ideal output: 0.242188
time: 29.700000, filtered data: 2.052734, difference with ideal output: 0.178711
time: 30.500000, filtered data: 2.143555, difference with ideal output: 0.172852
time: 31.300000, filtered data: 2.221680, difference with ideal output: 0.196289
time: 32.100000, filtered data: 2.260742, difference with ideal output: 0.228516
time: 32.900000, filtered data: 2.307617, difference with ideal output: 0.254883
time: 33.700000, filtered data: 2.364258, difference with ideal output: 0.236328
time: 34.500000, filtered data: 2.464844, difference with ideal output: 0.239258
```

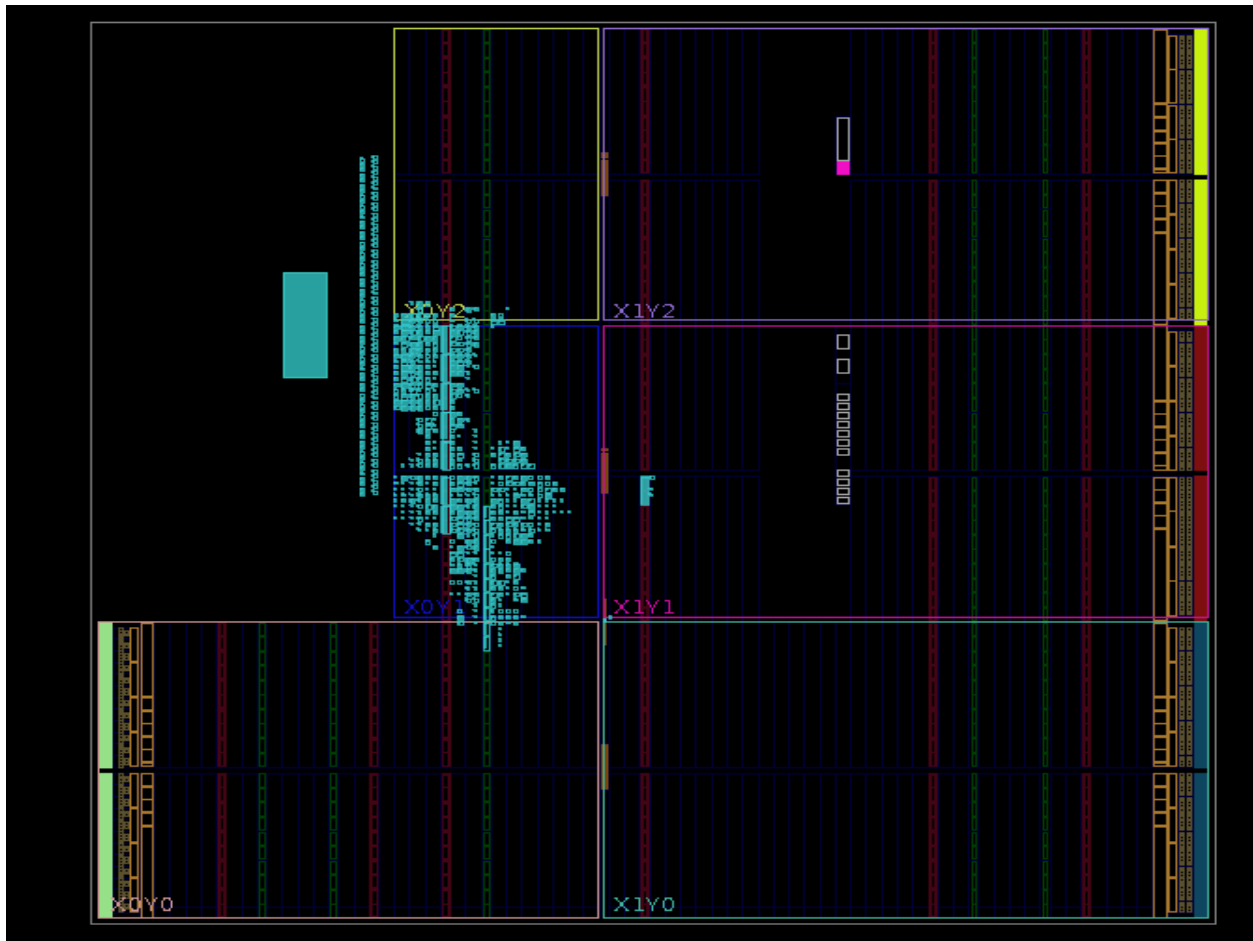
ort MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>


```
time: 32.900000, filtered data: 2.307617, difference with ideal output: 0.254883
time: 33.700000, filtered data: 2.364258, difference with ideal output: 0.236328
time: 34.500000, filtered data: 2.464844, difference with ideal output: 0.239258
time: 35.300000, filtered data: 2.577148, difference with ideal output: 0.296875
time: 36.100000, filtered data: 2.603516, difference with ideal output: 0.316406
time: 36.900000, filtered data: 2.640625, difference with ideal output: 0.333984
time: 37.700000, filtered data: 2.672852, difference with ideal output: 0.291016
time: 38.500000, filtered data: 2.742188, difference with ideal output: 0.262695
time: 39.300000, filtered data: 2.841797, difference with ideal output: 0.307617
time: 40.100000, filtered data: 2.878906, difference with ideal output: 0.337891
time: 40.900000, filtered data: 2.913086, difference with ideal output: 0.352539
time: 41.700000, filtered data: 2.965820, difference with ideal output: 0.329102
time: 42.500000, filtered data: 3.068359, difference with ideal output: 0.334961
time: 43.300000, filtered data: 3.141602, difference with ideal output: 0.353516
time: 44.100000, filtered data: 3.203125, difference with ideal output: 0.408203
time: 44.900000, filtered data: 3.238281, difference with ideal output: 0.423828
time: 45.700000, filtered data: 3.291992, difference with ideal output: 0.401367
time: 46.500000, filtered data: 3.399414, difference with ideal output: 0.412109
time: 47.300000, filtered data: 3.499023, difference with ideal output: 0.457031
time: 48.100000, filtered data: 3.560547, difference with ideal output: 0.511719
time: 48.900000, filtered data: 3.590820, difference with ideal output: 0.521484
time: 49.700000, filtered data: 3.647461, difference with ideal output: 0.502930
time: 50.500000, filtered data: 3.753906, difference with ideal output: 0.511719
time: 51.300000, filtered data: 3.834961, difference with ideal output: 0.539062
time: 52.100000, filtered data: 3.872070, difference with ideal output: 0.569336
time: 52.900000, filtered data: 3.913086, difference with ideal output: 0.589844
time: 53.700000, filtered data: 3.956055, difference with ideal output: 0.557617
time: 54.500000, filtered data: 4.042969, difference with ideal output: 0.546875
time: 55.300000, filtered data: 4.154297, difference with ideal output: 0.603516
time: 56.100000, filtered data: 4.179688, difference with ideal output: 0.622070
time: 56.900000, filtered data: 4.207031, difference with ideal output: 0.629883
time: 57.700000, filtered data: 4.233398, difference with ideal output: 0.580078
time: 58.500000, filtered data: 4.329102, difference with ideal output: 0.579102
time: 59.300000, filtered data: 4.392578, difference with ideal output: 0.587891
time: 60.100000, filtered data: 4.417969, difference with ideal output: 0.606445
time: 60.900000, filtered data: 4.454102, difference with ideal output: 0.623047
time: 61.700000, filtered data: 4.461914, difference with ideal output: 0.554688
time: 62.500000, filtered data: 4.522461, difference with ideal output: 0.518555
time: 63.300000, filtered data: 4.579102, difference with ideal output: 0.520508
time: 64.100000, filtered data: 4.607422, difference with ideal output: 0.541992
time: 64.900000, filtered data: 4.625977, difference with ideal output: 0.540039
time: 65.700000, filtered data: 4.678711, difference with ideal output: 0.517578
time: 66.500000, filtered data: 4.774414, difference with ideal output: 0.516602
time: 67.300000, filtered data: 4.869141, difference with ideal output: 0.556641
time: 68.100000, filtered data: 4.898438, difference with ideal output: 0.579102
time: 68.900000, filtered data: 4.944336, difference with ideal output: 0.604492
time: 69.700000, filtered data: 5.007812, difference with ideal output: 0.592773
time: 70.500000, filtered data: 5.085938, difference with ideal output: 0.573242
```

```
time: 71.300000, filtered data: 5.151367, difference with ideal output: 0.583984
time: 72.100000, filtered data: 5.196289, difference with ideal output: 0.622070
time: 72.900000, filtered data: 5.250000, difference with ideal output: 0.656250
time: 73.700000, filtered data: 5.291992, difference with ideal output: 0.623047
time: 74.500000, filtered data: 5.358398, difference with ideal output: 0.591797
time: 75.300000, filtered data: 5.463867, difference with ideal output: 0.642578
time: 76.100000, filtered data: 5.469727, difference with ideal output: 0.641602
time: 76.900000, filtered data: 5.505859, difference with ideal output: 0.658203
time: 77.700000, filtered data: 5.528320, difference with ideal output: 0.604492
time: 78.500000, filtered data: 5.601562, difference with ideal output: 0.581055
time: 79.300000, filtered data: 5.671875, difference with ideal output: 0.596680
time: 80.100000, filtered data: 5.722656, difference with ideal output: 0.640625
time: 80.900000, filtered data: 5.774414, difference with ideal output: 0.672852
time: 81.700000, filtered data: 5.834961, difference with ideal output: 0.657227
time: 82.500000, filtered data: 5.897461, difference with ideal output: 0.623047
time: 83.300000, filtered data: 5.990234, difference with ideal output: 0.661133
time: 84.100000, filtered data: 6.047852, difference with ideal output: 0.711914
time: 84.900000, filtered data: 6.116211, difference with ideal output: 0.759766
time: 85.700000, filtered data: 6.147461, difference with ideal output: 0.715820
time: 86.500000, filtered data: 6.239258, difference with ideal output: 0.709961
time: 87.300000, filtered data: 6.297852, difference with ideal output: 0.714844
time: 88.100000, filtered data: 6.357422, difference with ideal output: 0.767578
time: 88.900000, filtered data: 6.428711, difference with ideal output: 0.818359
time: 89.700000, filtered data: 6.440430, difference with ideal output: 0.754883
time: 90.500000, filtered data: 6.496094, difference with ideal output: 0.712891
time: 91.300000, filtered data: 6.598633, difference with ideal output: 0.760742
time: 92.100000, filtered data: 6.656250, difference with ideal output: 0.811523
time: 92.900000, filtered data: 6.690430, difference with ideal output: 0.826172
time: 93.700000, filtered data: 6.735352, difference with ideal output: 0.794922
time: 94.500000, filtered data: 6.820312, difference with ideal output: 0.783203
time: 95.300000, filtered data: 6.890625, difference with ideal output: 0.798828
time: 96.100000, filtered data: 6.936523, difference with ideal output: 0.837891
time: 96.900000, filtered data: 6.958984, difference with ideal output: 0.840820
time: 97.700000, filtered data: 6.979492, difference with ideal output: 0.785156
time: 98.500000, filtered data: 7.046875, difference with ideal output: 0.755859
```

zynq>

Implemented design:

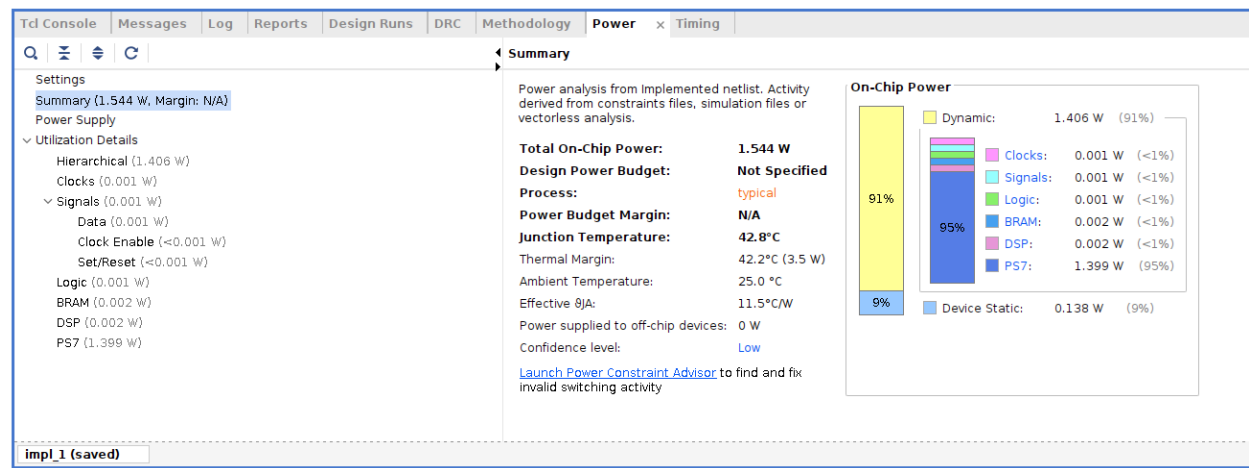


Timing summary:

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 26.945 ns	Worst Hold Slack (WHS): 0.071 ns	Worst Pulse Width Slack (WPWS): 15.520 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 1507	Total Number of Endpoints: 1507	Total Number of Endpoints: 663

All user specified timing constraints are met.

Power consumption :



Hardware Utilization :

DSP: Preliminary Mapping Report (see note below. The ' indicates corresponding REG is set)

Module Name	DSP Mapping	A Size	B Size	C Size	D Size	P Size	AREG	BREG	CREG	DREG	ADREG	MREG	PREG
multiply	A*B	18	10	-	-	48	0	0	-	-	-	0	0
multiply	A*B	18	18	-	-	48	0	0	-	-	-	0	0
multiply	A*B	18	10	-	-	48	0	0	-	-	-	0	0
multiply	A*B	18	18	-	-	48	0	0	-	-	-	0	0
kalman	A*B	16	16	-	-	32	0	0	-	-	-	0	0
multiply	A*B	18	10	-	-	48	0	0	-	-	-	0	0
multiply	A*B	18	18	-	-	48	0	0	-	-	-	0	0
multiply	A*B	18	10	-	-	48	0	0	-	-	-	0	0
multiply	A*B	18	18	-	-	48	0	0	-	-	-	0	0
kalman	A*B	16	16	-	-	32	0	0	-	-	-	0	0

Dynamic Shift Register Report:

Module Name	RTL Name	Length	Data Width	SRL16E	SRLC32E	Mux F7	Mux F8	Mux F9
dsrl	memory_reg[3]	20	20	20	0	0	0	0
dsrl_1	memory_reg[3]	2	2	2	0	0	0	0
dsrl_2	memory_reg[31]	34	34	0	34	0	0	0
dsrl_3	memory_reg[31]	13	13	0	13	0	0	0

Finished ROM, RAM, DSP, Shift Register and Retiming Reporting

Start Writing Synthesis Report

DSP Final Report (the ' indicates corresponding REG is set)

Module Name	DSP Mapping	A Size	B Size	C Size	D Size	P Size	AREG	BREG	CREG	DREG	ADREG	MREG	PREG
kalman	A*B'	16	16	-	-	32	1	1	-	-	-	0	0
kalman	A'*B'	16	16	-	-	32	1	1	-	-	-	0	0
multiply	A*B	17	9	-	-	48	0	0	-	-	-	0	0
multiply	A*B	17	17	-	-	48	0	0	-	-	-	0	0
multiply	A*B	17	9	-	-	48	0	0	-	-	-	0	0
multiply	A*B	17	17	-	-	48	0	0	-	-	-	0	0
multiply	A*B	17	9	-	-	48	0	0	-	-	-	0	0
multiply	A*B	17	17	-	-	48	0	0	-	-	-	0	0
multiply	A*B	17	9	-	-	48	0	0	-	-	-	0	0
multiply	A*B	17	17	-	-	48	0	0	-	-	-	0	0

Clock Summary:

The screenshot shows the 'Timing' window with the 'Clock Summary' tab selected. The table displays the following data:

Name	Waveform	Period (ns)	Frequency (MHz)
clk_fpga_0	{0.000 16.500}	33.000	30.303

Maximum Clock frequency:

From the post-implementation timing report, both **Total Negative Slack (TNS)** and **Worst Negative Slack (WNS)** were non-negative, confirming that the design met timing constraints and could be safely loaded onto the FPGA.

The **maximum operating frequency** of the design is calculated using the formula:

$$f_{\max} = 1 / [\text{Total clock period} - \text{Worst Negative Slack}]$$

$$f_{\max} = 1 / [33 - 26.945] = 165 \text{ MHz}$$

Thus, the design can operate reliably at a clock frequency of **165 MHz**.

Throughput calculation:

The maximum throughput is then estimated based on the clock frequency, data width, and number of cycles per output:

$$\text{Throughput} = f_{\text{max}} * [\text{Output bits} / \text{Clock Cycles per output}]$$

$$\text{Throughput} = 165 \text{ MHz} * [16/6] = 440 \text{ Mbps}$$

Hence, the implemented Kalman Filter achieves a maximum throughput of approximately **440 Mbps**, indicating that the FPGA design efficiently supports high-speed real-time filtering operations.

Contribution:

The Kalman Filter code was collaboratively developed by Sai Vikhyat Pattar, Pranav, and Premanshu as part of the Lab 7 assignment. Each member contributed to different stages of the design, coding, and testing process. The following section highlights my specific contribution to the project, detailing the portions of the code, debugging, and hardware verification tasks that I was responsible for:

Took primary responsibility for implementing the **main Kalman Filter module (kalman.v)**, developing the finite state machine (FSM) for the **IDLE, PREDICT, UPDATE, and OUTPUT_STATE** stages. Designed the control logic to coordinate prediction and update operations, managed input/output synchronization, and ensured accurate fixed-point arithmetic. Verified simulation results in **Vivado**, performed functional debugging, and finalized the hardware integration for FPGA testing. Also contributed to documenting the design methodology and analyzing results.

Inference:

The FPGA-based Kalman Filter successfully performed real-time state estimation with accurate convergence and stable operation. The design met timing constraints and achieved a maximum frequency of 165 MHz, resulting in an effective throughput of about 440 Mbps, demonstrating efficient hardware utilization and high processing speed.

Conclusion:

The Kalman Filter was efficiently implemented and verified on FPGA, achieving high performance and accuracy. The results confirmed correct functionality, stable estimation behavior, and excellent timing performance, highlighting the FPGA's capability for real-time signal processing and estimation applications.